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Robotics & Automation Laboratory

Final Project Report

Section: G2 Group: 04

Development of an Intelligent Mobile Manipulation System for
Daily Assistance of Visually Impaired Individuals

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1 Abstract

This project presents the development of an autonomous intelligent mobile manipulation system designed to assist visually impaired individuals with daily object handling tasks. The system integrates a Raspberry Pi 4B (8GB) processing unit, a 6-DOF robotic arm, and a dual-motor chassis for navigation. A hybrid perception approach is employed, combining an overhead camera for global workspace mapping and object localization using an ArUco marker network, with a front-facing Pi camera for precise object classification. The robot navigates to target coordinates, verifies object identity at a 32 cm threshold using an ultrasonic sonar, and executes a pick-and-place sequence.

2 Introduction

Visually impaired individuals face significant challenges in performing daily tasks that involve locating, identifying, and manipulating objects in their environment. This project addresses this complex engineering problem by developing an autonomous mobile manipulation system. The problem is complex due to the integration of multiple engineering domains, including mechatronics, computer vision, embedded systems, and control systems.

Alternative solutions to this problem include static assistive robots with a limited workspace, wearable haptic feedback devices, or purely software-based solutions. However, these alternatives often lack the physical manipulation capability or the mobile autonomy required for effective daily assistance.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project is to design and prototype an autonomous mobile manipulation system aimed at assisting visually impaired individuals in everyday object handling tasks. The system must be capable of operating in a structured tabletop environment, where it can autonomously navigate, identify objects, and manipulate them using a 6-DOF robotic arm. The system will need to reliably select objects based on shape, color, or other distinguishing features, move them to a designated target location, and return to the starting point for the next task. The design also requires an efficient perception system combining overhead vision and local sensing for object verification and precise movement. The project will consider cost-effectiveness and usability for visually impaired users.

3.1.2 Literature Review

Research in the field of assistive robotics has evolved with advancements in both mobile manipulation and assistive technologies for people with disabilities. Studies on mobile manipulators for the visually impaired include systems using technologies like LIDAR, SLAM, and robotic arms for object manipulation. Notably, systems like the Robotic Assistant for the Visually Impaired leverage LIDAR and SLAM algorithms for navigation but may struggle with object manipulation due to limited precision in smaller, cluttered environments. Alternatively, vision-based systems utilizing fiducial markers, like the ArUco marker system, have shown great potential for high-precision localization and object detection in controlled settings. These systems offer a low-cost, modular approach to achieving real-time localization, making them a promising solution for indoor assistive environments. The review of current literature indicates that vision-based hybrid systems combining overhead and local vision can potentially optimize task accuracy while reducing computational overhead in real-time mobile robotics.

3.1.3 Formulation of Problem

The problem this project addresses is the development of an autonomous assistive robotic system capable of performing pick-and-place operations for visually impaired users in a tabletop environment. The system must autonomously:

1. Navigate the workspace using a dual-camera vision system to establish an overhead global frame and local object verification.
2. Identify objects by recognizing specific shapes, colors, or patterns using machine learning models and verify the object using front-facing vision.
3. Manipulate objects with precision using a 6-DOF robotic arm, placing them at a specified target marker location.
4. The system must also be capable of operating autonomously in a low-speed, low-risk environment while ensuring high reliability and safe interaction with objects. The core challenge is to minimize system error across perception, decision-making, and actuation phases, ensuring that the robot can perform the task with minimal human intervention. Furthermore, the system should be scalable and adaptable for future use beyond the tabletop environment.

3.1.4 Analysis

The analysis for the problem formulation involved breaking down the project into distinct functional modules:

- Global localization: Using ArUco markers to map the workspace and define robot position.
- Local object verification: Using the front-facing Pi Camera for close-up object classification based on shape or color.
- Navigation and precision control: Calculating the Euclidean distance between the robot and target locations, with sonar-based stopping logic to ensure precise object handling.
- Manipulation: Using a 6-DOF robotic arm, with servo-driven motion based on pre-programmed trajectories to achieve object grasping and placement.

The system must account for error tolerances in each of these modules. For instance, the sonar sensor has an error tolerance of ± 1 cm, while the 6-DOF arm is limited in torque and capable of manipulating only lightweight objects. Additionally, external factors such as lighting conditions and camera resolution can affect the performance of marker detection and object classification. To mitigate errors, the system uses a threshold-based control logic that fine-tunes the movement parameters and integrates feedback for precision positioning. A key consideration in the analysis was the trade-off between computational load and real-time task execution. The use of hybrid perception, with overhead vision and local sensors, effectively reduces the computational demand and improves the overall task reliability.

3.2 Design Method (PO(a))

The design methodology for this project involved a systematic approach that integrates concepts from mathematics, computer science, and control systems to develop an autonomous robotic manipulation system. The approach focuses on perception, navigation, and manipulation, with particular attention to reducing computational overhead and ensuring precision in object handling. Below is the breakdown of the methods used in each domain:

- **Mathematics (Coordinate Geometry):** The system uses Euclidean distance to calculate the shortest path to a target and to determine the robot's position relative to objects using overhead vision and ArUco markers. This provides the system with a reliable spatial understanding of the environment.
- **Computer Science (Machine Learning and Image Processing):** Two models were trained for object classification. Model 1 was used for global perception via the overhead camera, focusing on object localization. Model 2 processed the close-up images from the front-facing Pi camera to identify specific objects based on shape and color.
- **Control Systems:** A threshold-based control system was implemented for movement, using an ultrasonic sensor to ensure the robot stops within an acceptable range (32 cm from the object). This is further complemented by motion control algorithms that adjust robot movement in real-time based on sensory feedback.
- The design method effectively integrates theory and practical applications, ensuring the system is both efficient and adaptable to different assistive scenarios.

3.3 Hardware Design

The hardware design for the **autonomous mobile manipulator** consists of several interconnected components, each playing a vital role in enabling the system to perform its tasks efficiently. These components include **sensors, actuators, processing units, and communication interfaces**.

Key components of the hardware design are as follows:

- **Raspberry Pi 4B (8GB):** This serves as the **central processing unit** for the entire system, running the control algorithms, machine learning models, and handling the communication between the sensors, actuators, and the robot.
- **6-DOF Robotic Arm:** The **manipulation system** is based on a six-degree-of-freedom robotic arm, which is capable of **precise object grasping and placement**. The arm is actuated using **high-torque servos** controlled by the **PCA9685 PWM servo driver**.
- **Dual Motor Chassis:** The **mobile platform** is powered by a dual-motor chassis that enables the robot to navigate autonomously within a defined workspace. The motors are driven by the **L298N motor driver**.
- **Cameras:** The system uses two cameras:
 - **Overhead Camera (USB Camera):** Provides a **top-down view** for **global localization**, object detection, and mapping of the workspace using **ArUco markers**.
 - **Pi Camera (Front-facing):** Captures detailed, **close-up images** of objects for **object classification**.
- **Ultrasonic Sensor (HC-SR04):** Provides precise **distance measurements** to assist in **collision avoidance** and **precision stopping** before object manipulation.

- **Power Supply:** The robot is powered by a **12V rechargeable battery pack**, ensuring sufficient power for all components. The **Raspberry Pi** is powered through a regulated power supply, while the motors and servos receive power from the battery.

The interconnections of these components are as follows:

1. The **overhead camera** is mounted to the top of the workspace and connected to the **Raspberry Pi** via USB.
2. The **Pi Camera** is mounted on the front of the robot and connected to the **Raspberry Pi** for image classification.
3. The **PCA9685 servo driver** controls the servos in the **6-DOF arm**.
4. The **L298N motor driver** interfaces with the **dual motor chassis** for movement control.
5. The **HC-SR04 ultrasonic sensor** is used to measure the distance between the robot and objects in the environment, providing feedback for precision stopping.

4 Implementation

4.1 Description

The implementation phase focuses on the integration of all subsystems to create a cohesive, functional robot. This phase begins with setting up the hardware and establishing the communication protocols between components, followed by the installation of software for object detection, classification, and manipulation. The implementation process is broken down into the following steps:

1. **Hardware Integration:** The first step was to assemble and integrate all hardware components, ensuring proper connections between the Raspberry Pi, sensors, actuators, and cameras. This required careful wiring, mounting of sensors and cameras, and configuring the motor and servo drivers.
2. **System Initialization:** After the hardware setup, the system was initialized with the Raspberry Pi running the operating system and necessary libraries. The Raspberry Pi OS was configured, and the device drivers for the motor driver, servo controller, and sensors were installed.
3. **Perception System Setup:**
 - **Global Localization:** The overhead camera was calibrated to capture a top-down view of the workspace. ArUco marker detection was implemented to establish a coordinate system, allowing the robot to map its environment and locate itself relative to the defined workspace.
 - **Object Classification:** The front-facing Pi Camera was used to capture high-resolution images of the objects. A machine learning model was trained to classify objects based on shape and color. The model was deployed on the Raspberry Pi, enabling real-time object classification.
4. **Navigation and Precision Control:** The robot's movement was controlled using the dual-motor chassis. A distance threshold system was implemented using the HC-SR04 ultrasonic sensor to stop the robot when it approaches an object within a predefined distance (32 cm). A feedback loop was created to ensure the robot stays within the correct path and distance from the objects.
5. **Robotic Arm Manipulation:** The 6-DOF arm was calibrated to follow pre-programmed trajectories based on object positioning data. The system used servo motors to control the arm, and the PCA9685 driver provided precise movement control. Once an object was identified, the arm would execute the pick-up and release actions.
6. **Testing and Debugging:** After the system was fully assembled and programmed, rigorous testing was conducted to ensure that the robot could reliably detect, navigate, and manipulate objects. The precision of stopping and object handling was validated through multiple test cases.

7. **Final System Integration:** The final step involved combining all modules (perception, navigation, manipulation) into a seamless, integrated system. The robot was tested in real-world conditions to ensure smooth operation and task accuracy.

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of the proposed system lies in its integration of mobile navigation, dual-layer vision, and robotic manipulation within a single assistive platform. Unlike conventional systems that depend on only one sensing modality, this project combines global overhead vision for workspace mapping and object localization with local front-facing vision for close-range object verification. This hybrid approach reduces computational burden, improves positioning reliability, and supports more accurate pick-and-place operation in a structured indoor environment.

5.2 Design Considerations (PO(c)).

5.2.1 Considerations to public health and safety

Public health and safety were considered by designing the system for low-speed operation in a controlled tabletop environment. The robotic arm handles only lightweight objects, and the stopping logic based on ultrasonic sensing reduces the chance of collision. The system also uses a defined workspace with ArUco markers, which improves predictability and operational safety during navigation and manipulation.

5.2.2 Considerations to environment

Environmental considerations were addressed through the use of low-power embedded hardware, rechargeable power sources, and a compact prototype design. The project avoids excessive material use by integrating commercially available modular components. In addition, local processing on the Raspberry Pi reduces dependency on external cloud computation, thereby minimizing communication overhead and energy demand.

5.2.3 Considerations to cultural and societal needs

The project addresses an important societal need by supporting greater independence for visually impaired individuals in performing everyday object-handling tasks. The design focuses on usability in familiar indoor settings and aims to reduce reliance on constant human assistance. As a result, the system has potential value in improving accessibility, confidence, and quality of life for users requiring assistive support.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The experimental evaluation was designed to verify four key functions of the system: workspace mapping, target localization, precision stopping, and object manipulation. The robot was tested in a bounded tabletop setup with ArUco markers, scattered colored objects, and a fixed destination marker. The experiments focused on whether the system could correctly identify the nearest target, stop at the desired operational distance, verify the object using the front-facing camera, and complete the pick-and-place sequence.

5.3.2 Data Collection

Data were collected from multiple subsystems during implementation and testing. The overhead camera provided object coordinates and robot position information, while the ultrasonic sensor supplied distance measurements for precision stopping. The front-facing camera captured close-range object images for classification, and the robotic arm response was observed during grasping and release operations. Performance observations from repeated trials were used to assess system consistency and reliability.

5.4 Limitations of Tools (PO(e))

- Raspberry Pi 4B: Limited for running two real-time ML models simultaneously
- HC-SR04 Sonar: Affected by surface texture, $\pm 1\text{-}2$ cm errors possible
- ArUco Marker Detection: Requires good lighting conditions
- 6-DOF Arm: Limited torque restricts to lightweight objects only

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

From a societal perspective, the project has positive impact because it aims to improve accessibility and reduce dependency for visually impaired persons. Culturally, the system supports inclusive engineering by focusing on a user group whose needs are often underrepresented in low-cost robotics research. The design also encourages the development of assistive technologies that are practical for local and resource-constrained settings.

5.5.2 Assessment of Health and Safety Issues

Health and safety issues were considered through controlled motion, limited operating speed, and sensor-based stopping logic. Since the robot interacts physically with objects, maintaining safe movement and predictable arm motion is essential. The tabletop setup, low-force actuation, and defined workspace all reduce operational risks during testing and demonstration.

5.5.3 Assessment of Legal Issues

The legal issues relevant to this project mainly relate to safe deployment, intellectual property, and data handling. Any future real-world implementation should ensure compliance with safety standards for robotic devices and proper attribution of external libraries, datasets, and software tools. If user-related video or environmental data are stored in future versions, clear consent and privacy protection policies must also be maintained.

5.6 Sustainability Evaluation (PO(g))

The sustainability of the project lies in its modular, repairable, and upgradeable design. Since the system uses widely available components, faulty modules can be replaced without redesigning the entire prototype. The use of rechargeable power and low-power processing also supports energy-efficient operation. Furthermore, the system provides a foundation for future low-cost assistive solutions, which contributes to the broader goal of sustainable and inclusive technological development.

5.7 Ethical Issues (PO(h))

Ethical principles were applied throughout the project by prioritizing beneficence, safety, privacy, and responsible design. The main goal of the system is to support visually impaired individuals in a way that promotes independence and dignity. One ethical challenge was balancing full autonomy with user safety. This was mitigated by restricting the robot to a controlled workspace, limiting speed, and introducing distance-based stopping before arm motion. Another concern was privacy in vision-based operation. To reduce this issue, the system processes information locally on the Raspberry Pi rather than depending on external cloud services. The team also maintained academic honesty by documenting contributions and using a structured engineering workflow.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

ID	Contribution
2006099	Raspberry Pi OS Installation, Motor Calibration , ARM Setup, Hardware setup
2006102	ARM setup, Hardware, Navigation, Model Training, Motor Calibration, Sonar Integration and calibration
2006112	Raspberry Pi OS Installation, Motor Calibration, Sonar Integration and calibration, Navigation, Precision Control
2006117	Model Training ,Labeling, Precision Control, Hardware Setup
2006118	Dataset for front face Model training, Hardware setup, Navigation, Precision Control
2006137	Hardware setup, Dataset for front face Model training ,ARM setup, Motor Calibration,Alghoritm

6.2 Mode of TeamWork

The team followed a collaborative and task-oriented workflow. Responsibilities were distributed according to technical strengths, with parallel work carried out on hardware setup, machine learning, navigation, and calibration. Regular discussion and integration sessions were necessary to ensure compatibility between sensing, decision logic, and actuation. Final testing and refinement were conducted jointly to evaluate full-system performance.

6.3 Diversity Statement of Team

The team demonstrated diversity in terms of technical skills, problem-solving approaches, and areas of contribution. Different members focused on hardware, software, control, dataset preparation, and model-related tasks. This diversity improved the quality of the final outcome by allowing the project to be approached from multiple engineering perspectives and by supporting effective division of responsibilities.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role
Week 1	Project topic selection and problem identification	All members discussed the project idea and identified the main problem to be solved	Defined the overall project objective and scope
Week 2	System planning and component selection	Members reviewed possible hardware and software components for implementation	Finalized system architecture and selected required components
Week 3	Procurement and hardware collection	Members collected Raspberry Pi, robotic arm, motor driver, cameras, sonar sensor, and chassis components	Ensured availability of all major components
Week 4	Raspberry Pi setup and initial hardware integration	Assigned members installed OS, configured Raspberry Pi, and checked peripheral connections	Verified basic system startup and hardware response
Week 5	Chassis assembly and motor	Members assembled the mobile platform and calibrated the drive motors	Tested basic forward, backward, and turning motion
Week 6	Robotic arm setup and servo calibration	Assigned members mounted the arm and calibrated servo angles for stable movement	Verified arm motion and gripping sequence
Week 7	Overhead camera setup and ArUco marker mapping	Vision team configured the overhead camera and implemented marker-based coordinate mapping	Established workspace boundary and robot localization
Week 8	Front-facing camera dataset preparation and model training	ML team collected data, labeled images, and trained the object classification model	Reviewed training results and prepared model for deployment

Week 9	Sonar integration and precision stopping logic	Members integrated the ultrasonic sensor and tuned the stopping threshold	Tested object distance measurement and stopping response
Week 10	Navigation algorithm development	Assigned members implemented target selection and movement logic	Coordinated navigation with perception outputs
Week 11	Full system integration	All members connected perception, navigation, sonar, and arm subsystems	Tested complete operational workflow
Week 12	Final testing, evaluation, and report preparation	Members conducted repeated trials, recorded observations, and prepared documentation	Evaluated system performance and finalized project report

7 Communication to External Stakeholders (PO(j)).

7.1 Executive Summary

A team of BUET students has developed an intelligent mobile robot designed to assist visually impaired individuals with everyday object-handling tasks. The system combines camera-based mapping, ultrasonic sensing, and a six-degree-of-freedom robotic arm to find, identify, pick, and place selected objects automatically. An overhead camera locates objects and tracks the robot, while a front-facing camera confirms the target before grasping. By integrating navigation, object recognition, and manipulation in a single platform, the project demonstrates a practical low-cost assistive technology concept that can improve independence, safety, and convenience in structured indoor environments for users who require reliable assistance during daily tasks.

7.2 User Manual

Step 1: Prepare the workspace.

Place the four corner ArUco markers at the corners of the table. Put the destination marker in its fixed position and place the target objects inside the workspace.

Step 2: Set up the hardware.

Position the overhead camera above the workspace so that the full table is visible. Mount and connect the front-facing Pi Camera, ultrasonic sensor, robotic arm, motor driver, and power supply properly.

Step 3: Power on the system.

Turn on the Raspberry Pi, motor drivers, cameras, and robotic platform. Ensure that all components are connected and initialized correctly.

Step 4: Start the program.

Run the main control program from the Raspberry Pi. The system will begin by detecting the ArUco markers and generating the workspace coordinate system.

Step 5: Detect and select target.

The overhead camera will identify the robot position and the object positions. The control logic will select the nearest valid target coordinate.

Step 6: Autonomous navigation.

The mobile platform will move toward the selected target while maintaining its heading and monitoring position.

Step 7: Precision stopping.

When the ultrasonic sensor detects the object at the predefined threshold, the robot will stop and hold position for stable front-view inspection.

Step 8: Local object verification.

The front-facing camera will capture a close image of the object. The classification model will determine whether the object matches the desired target criteria.

Step 9: Grasping operation.

If the object is valid, the 6-DOF robotic arm will execute the programmed pick-up sequence. If the object is invalid, the system will skip it and move to the next target.

Step 10: Delivery operation.

After grasping the object, the robot will navigate to the destination marker and execute the release sequence.

Step 11: Shutdown procedure.

After task completion, stop the control program and safely disconnect the power supply.

Step 12: Safety precautions.

Operate the robot only in a clear and controlled environment. Avoid placing heavy or fragile objects in the test area. Keep hands away from the robotic arm during movement.

8 Project Management and Cost Analysis (PO(k))

Name of Components	date	cost
1.Pi(8 GB)+ Casing+Heat Sink+Adapter+transport	20/01/26	12200+200
2.pi camera+transport	31/01/26	514.25+130
3.SD card+Card reader+Box	3/2/2026	530+180
4.Stick Gun+Cutter	4/2/2026	380+40
5.Buet to Patuatuli to Nawab. To buet transport	18/01/2026	330
7.Arm setup from robotics bd+Transport	4/1/2026	7336+630
8.Car Set up		7885
9.Clamper from Nobabpur	2/4/2026	600+60
10.Equipment+Glue Stick	3/4/2026	408+40
11.soanr+rickshaw	3/4/2026	290
12. 2 ta battery casing +wire	3/4/1936	100
14.Memory(32 gB)	4/4/2026	120+480
15.Li- ion Battery	5/ 4/2026	530
18.Wasif		390
Total Cost		33373.25

9 Future Work (PO(I))

Future work may include expanding the system beyond tabletop operation to larger indoor environments, integrating SLAM-based navigation for dynamic spaces, adding voice-command interaction for improved accessibility, and incorporating adaptive grasp control using force or tactile feedback. The perception system can also be improved through stronger object recognition models and better robustness under varying lighting conditions. In addition, future versions may focus on reducing hardware cost and improving portability for practical deployment in real assistive settings.

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